

Week 2 covers sections of chapter 14 in the textbook. Topics include:

- heat, energy, and power
- heat capacity, specific heat capacity, and molar heat capacity
- latent heat of fusion or vaporization

1. Suppose you have 2 kg of water in a cup and you put it in the microwave. What form of heating is this (conduction, convection, or radiation)? It takes 1 minute to heat up from 20 °C to 40 °C. What is the temperature change in Celsius (ΔT)? What is the change in temperature in Kelvin? Look up the specific heat of water and be careful to specify the units. How much heat would it take to accomplish this temperature change?

$$Q = m \cdot c \cdot \Delta T$$

↑ ↑
mass specific heat

← change in T

2. What are the units of power? What is another way to express those units? To follow up the previous problem, what is the power delivered to the water by the microwave? If this rate of heat delivery continues, how long will it take for the water to reach its boiling point?

3. You have three samples of material, all 1 kg each. The materials are gold, copper, and aluminum. Put these in order of how much heat is necessary to change the temperature 1 degree.

4. You have the same materials as in the last problem. If you put the same amount of heat into each sample, put them in order of which would heat up the most (remember that when we say "heat up" we mean increase in temperature).

5. How much heat does it take to bring 2 kg of aluminum from 25°C to 50°C ?

6. If it takes 5000 J to bring an ingot of gold from 25°C to 50°C , then what mass of gold is the ingot?

7. If 5000 J of heat goes into a 1.5 kg glass dish that was initially at 20°C , then what is its final temperature?

8. If a 5 kg block of aluminum has a temperature of 500°C , how much heat does it give off to cool down to 490°C ? What should the sign of heat be in this case?

9. If the aluminum block in the above problem gave up this heat because it was put in contact with a cooler 5 kg block of lead, then by how much does the temperature of the lead rise? If the lead was originally 75°C , then what would be its new temperature?

10. Keep this process going, the aluminum cooling off by 10 degrees and the lead warming up by whatever you found in the above problem. Approximately what temperature would they meet? This temperature is known as the *equilibrium temperature* since

after this heat flow stops, and they remain at the same temperature (ignoring heat that they lose to the surroundings).

11. Lets do this in one step now. If the aluminum has an initial temperature of 500°C and an *unknown final temperature*, and the lead starts at 75°C and has an unknown final temperature but the same final temperature as the aluminum, since that is the equilibrium temperature, then how can we find this with one expression? (*Hint: the overall energy of the system does not change. So any change in energy of one plus the change in energy of the other must be zero.*)

12. The exact same logic that applied to the above problem, applies to mixing two substances together. You can still treat them as separate substances with one or more giving up energy in the form of heat to the others. It is always assumed that the materials are kept in a well insulated container so that no heat is lost to the surroundings. This is a good way of measuring the heat capacity of an unknown material by mixing it with a material of known initial temperature and heat capacity and then measuring the equilibrium temperature that results. So suppose you start with 100 g of water at 90°C , and you pour in 50 g of unknown metal at 20°C . You stir the mixture and notice that the temperature of the mixture comes down to 40°C and

- How much work will it take for her to reach this height?
- Assuming that she is only 25% efficient at converting chemical energy from food into mechanical work, and that essentially all of the mechanical work is used to climb vertically, roughly how many bowls of corn flakes should the hiker eat before setting out? (standard serving size 1 oz, 100 Calories)
- As the hiker climbs the mountain the other 75% of the energy from the corn flakes is converted into thermal energy. If there were no way to dissipate this energy, by how many degrees would her body temperature increase? (Assume the human body is mostly water so that it has the same specific heat as water.)

- In fact, the extra energy does not warm the hiker's body significantly; instead, it goes (mostly) into evaporating water from her skin and within her lungs. How many liters of water should she drink during the hike to replace the lost fluids? (At 25°C , a reasonable outdoor temperature to assume, the latent heat of vaporization of water is 580 cal/g , 8% more than at 100°C .)

